VEHICLE SPEED–TIME ANALYSIS AND ESTIMATION OF MOTOR & INVERTER POWER

Tata Nexon.ev reference vehicle | MATLAB/Simulink drive-cycle simulation

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Abstract

This assignment uses the supplied urban drive-cycle data to estimate road-load forces and motor/inverter power for a Tata Nexon.ev reference vehicle. The Excel speed samples are plotted in MATLAB and used by a simple Simulink model to calculate rolling resistance, aerodynamic drag, acceleration force, total tractive force, motor mechanical power, and inverter DC input power. The two submitted simulation output graphs are included and discussed.

1. Aim and Objectives

Aim: To use speed-versus-time data to estimate vehicle forces and power requirements, and to observe the results in MATLAB/Simulink.

Objectives: (1) import speed-time data; (2) convert km/h to m/s; (3) estimate acceleration; (4) calculate road-load forces; (5) estimate wheel, motor and inverter power; (6) plot and interpret the simulation outputs.

2. Drive-Cycle Data

The six measurement points supplied in Drive_Cycle.xlsx are listed below. A start point at 0 s and 0 km/h was added for the simulation; this is an assumption and should be used only if the vehicle began from rest.

Time (s)	Speed (km/h)	Speed (m/s)
0	0	0.000
60	6	1.667
120	20	5.556
180	29	8.056
240	50	13.889
300	31	8.611
360	8	2.222

The measurements are spaced 60 seconds apart. Simulink interpolates between points, so brief acceleration and braking events between samples are not represented.

3. Vehicle Parameters and Assumptions

Parameter	Symbol	Value
Vehicle mass	m	1400 kg (provisional)
Gravity	g	9.81 m/s²
Rolling resistance coefficient	Crr	0.011 (assumed)
Drag coefficient	Cd	0.29 (assumed)
Frontal area	A	2.35 m² (assumed)
Air density	rho	1.225 kg/m³
Drivetrain efficiency	eta_d	0.90 (assumed)
Motor efficiency	eta_m	0.92 (assumed)
Inverter efficiency	eta_inv	0.97 (assumed)
Road gradient	theta	0° (flat road)

Mass and aerodynamic parameters are provisional model assumptions, not all verified manufacturer-published specifications for the exact Nexon.ev variant.

4. Governing Equations

- Speed: v (m/s) = speed (km/h) / 3.6
- Acceleration: a = (v_next - v_previous) / (t_next - t_previous)
- Rolling resistance: F_roll = Crr × m × g × cos(theta)
- Aerodynamic drag: F_drag = 0.5 × rho × Cd × A × v²
- Acceleration force: F_acc = m × a
- Gradient force: F_grade = m × g × sin(theta); flat road gives zero
- Total tractive force: F_total = F_roll + F_drag + F_acc + F_grade
- Wheel power: P_wheel = F_total × v
- Motor shaft power: P_motor = max(P_wheel, 0) / eta_drive
- Inverter DC input power: P_inv = P_motor / (eta_motor × eta_inv)

Forces are in newtons (N), speed in m/s, acceleration in m/s², and power in watts (W). Divide watts by 1000 to obtain kW. The model clips negative wheel power to zero; regenerative braking is not included.

5. Sample Calculation at 50 km/h

• 1) Convert speed: v = 50 / 3.6 = 13.889 m/s.

• 2) Rolling force: F_roll = 0.011 × 1400 × 9.81 = 151.07 N.

• 3) Aerodynamic drag: F_drag = 0.5 × 1.225 × 0.29 × 2.35 × (13.889)² = 80.52 N.

• 4) At constant speed, acceleration is zero: F_acc = 0 N. Flat-road gradient force = 0 N.

• 5) Total force: F_total = 151.07 + 80.52 = 231.59 N.

• 6) Wheel power: P_wheel = 231.59 × 13.889 = 3.22 kW.

• 7) Motor shaft power: P_motor = 3.22 / 0.90 = 3.57 kW.

• 8) Inverter DC input power: P_inv = 3.57 / (0.92 × 0.97) = 4.00 kW.

This is a steady-speed reference calculation. The instantaneous values in the simulation differ when acceleration is non-zero.

6. Simulation Outputs

Figure 1 shows the speed-time curve from the Excel measurements. The maximum speed is 50 km/h at 240 s, followed by a decrease to 31 km/h at 300 s and 8 km/h at 360 s.

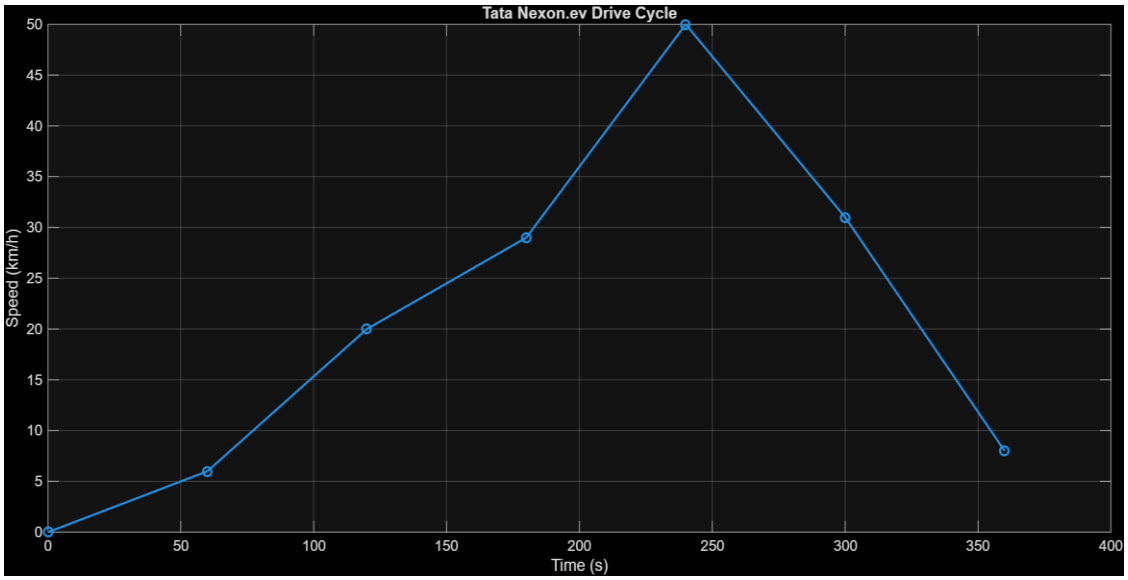


Figure 1. Speed versus time obtained in MATLAB.

Figure 2 shows the force and power outputs from the Simulink model. Rolling resistance is nearly constant, aerodynamic drag increases with speed squared, and acceleration force becomes negative during deceleration. Motor and inverter power are positive during motoring because negative wheel power is clipped to zero.

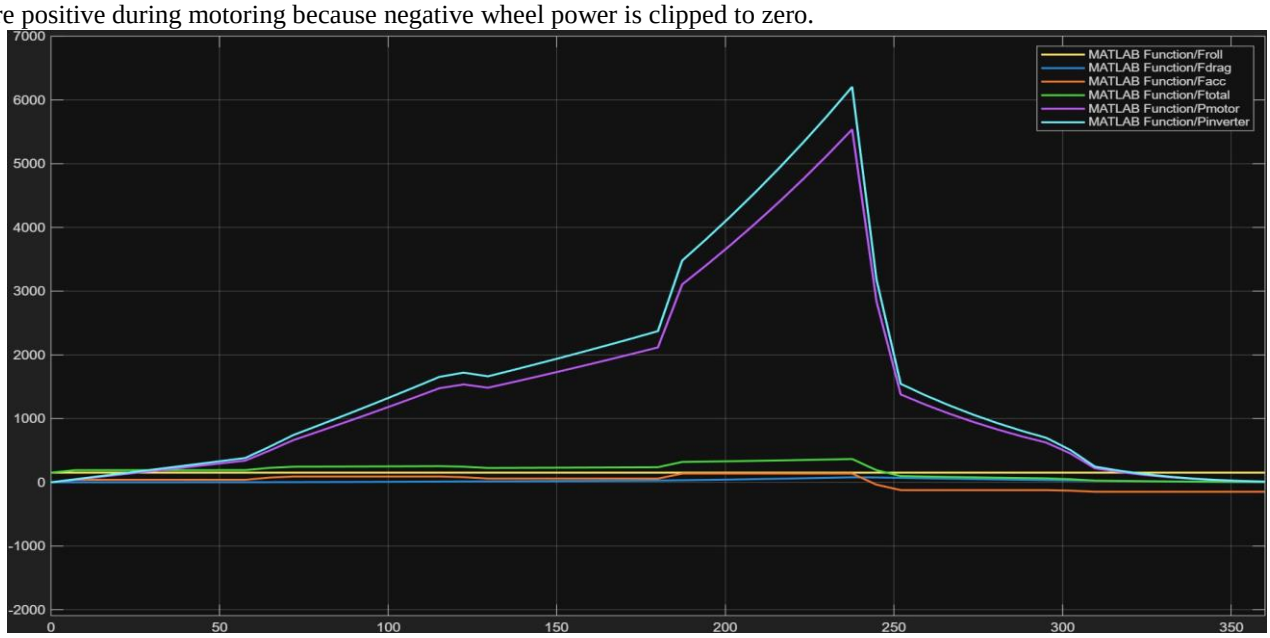


Figure 2. Simulink force and power output traces. From the displayed plot, peak motor power is approximately 5.5 kW and peak inverter DC input power approximately 6.2 kW.

7. Discussion

The drive cycle represents stop-start urban driving. At higher speed, aerodynamic drag contributes more significantly because it varies with v^2. Acceleration force depends on the gradient of the speed-time curve. The inverter input power is higher than motor mechanical power because the assumed motor and inverter efficiencies are less than 100%. The graph indicates an approximate peak of 5.5 kW motor power and 6.2 kW inverter input power for this simulated profile.

These peaks are simulation estimates for the sampled/interpolated drive cycle, not the rated power of the production vehicle. Because speed samples are 60 seconds apart, the model may miss short power peaks.

8. Limitations

• The data contains only six speed samples over 360 seconds.
• The starting point at 0 s and 0 km/h is assumed.
• Vehicle mass, drag coefficient, frontal area, rolling resistance and efficiencies are provisional assumptions.
• The road is assumed flat; wind, auxiliary loads, battery voltage sag, thermal limits, inverter switching losses and regenerative braking are not modelled.

9. Conclusion

The speed-time data was used to estimate vehicle road-load forces and electrical power demand. At a constant 50 km/h on a flat road, the model estimates 231.59 N total tractive force, 3.22 kW wheel power, 3.57 kW motor shaft power and 4.00 kW inverter DC input power. The two Simulink plots show the relationship between the urban speed profile, forces and power. More frequent speed measurements and verified vehicle parameters would improve peak-power accuracy.

Appendix: Simulink block flow

From Workspace (speed_ts) □ Derivative (acceleration) □ MATLAB Function (force and power calculations) □ Mux (six outputs) □ Scope. Connect the speed signal directly to the MATLAB Function and a speed Scope as well.